

SmartHome Plate

Smart Strike Zone Trainer & Pitching Analytics

Complete Build Guide

Design Specification & Engineering Analysis

Bill of Materials & Assembly Instructions

Firmware Architecture & App Wireframes

WITHSTANDS 105+ MPH BASEBALL IMPACTS

Dual Mode: Pitching Training + Hitting Analytics

Auto-Adjusting Strike Zone | Spin Rate | Pitch Movement | Exit Velocity

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For Prototype Build

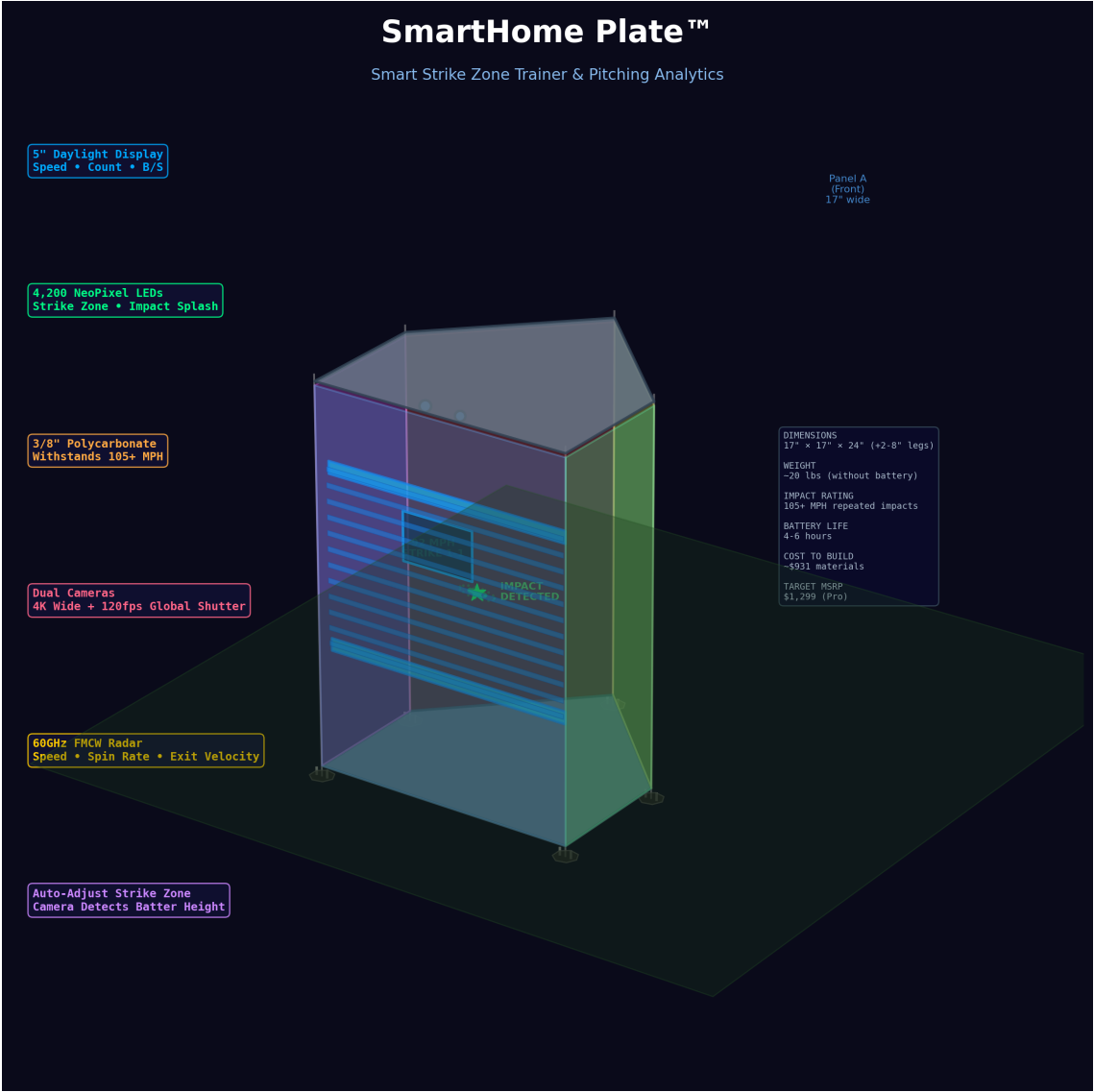


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1. Product Overview & Features

SmartHome Plate is a ruggedized, impact-rated smart device shaped like official home plate that sits on top of a standard baseball home plate. It provides real-time pitch tracking, instant HD video replay, radar speed measurement, spin rate analysis, pitch movement quantification, and arcade-style training games - all in a housing that survives repeated 105+ mph baseball impacts.

Primary Functions

- Strike Zone Trainer - Detects ball/strike contact on any of five vertical faces
- HD Recording - Wide-angle (4K/30fps) + high-speed (120fps global shutter) cameras
- Radar Speed - 60GHz FMCW Doppler measures pitch velocity (40-110 mph)
- Spin Rate Detection - Micro-Doppler signatures from radar analyze spin (0-3500 RPM)
- Pitch Movement Analysis - Tracks curveball break, slider sweep, knuckleball dance
- Pitch Type Classification - ML auto-classifies fastball, curveball, slider, changeup, etc.
- Exit Velocity Tracking - Measures ball speed off the bat (hitting mode)
- Launch Angle Detection - Calculates ball departure angle after bat contact
- Batter Swing Analysis - Camera tracks bat path, swing plane, late/early detection
- Auto-Adjusting Strike Zone - Camera ML detects batter height and sets zone
- Live Display - 5" daylight-readable screen shows speed, count, spin, movement
- LED Feedback - 4,200+ NeoPixels illuminate precisely where ball contacts the panel
- Arcade Games - Simon Says, Pop-a-Shot, competitive pitch count games
- Companion App - iOS/Android for session review, stats, leaderboards

Target Users

- Competitive pitchers (Little League through Pro)
- Pitching coaches and academies
- Hitters and batting coaches (exit velocity + launch angle)
- Training facilities and batting cages
- Backyard/family use
- Scouting and player evaluation

Key Specifications

Parameter	Value
Footprint	Official home plate (17" front x 17" depth)
Body Height	24" (adjustable to 26", 29", 32" with legs)
Weight	~20 lbs (without battery), ~23 lbs (with)
Impact Rating	Withstands 105+ mph repeated baseball impacts
Panels	3/8" polycarbonate (Lexan), UV-hardcoated
Frame	1/4" 6061-T6 aluminum, bead-blasted, anodized
Display	5" IPS, 800x480, 1000 nits daylight-readable
Cameras	4K wide-angle + 120fps global shutter

Radar	60GHz FMCW (speed, spin rate, movement)
LEDs	~4,200 WS2812B NeoPixels, individually addressable
Impact Detection	20x piezo film sensors, acoustic triangulation
Battery	6x 21700 Li-Ion (111Wh), 4-6 hour runtime
Wireless	WiFi 6 + Bluetooth 5.2
Compute	Jetson Orin Nano (Pro) or RPi CM4 (Standard)
Weatherproof	IP54 (rain resistant)

2. Design Specification

Home Plate Geometry

The device matches official MLB home plate dimensions - a 17-inch wide pentagon with two 8.5-inch sides, two 12-inch diagonal sides, and a rear point facing the catcher.

Vertices (mm, origin at center of front edge)

Vertex	Coordinates (X, Y)
V1 (Front-Left)	(-215.9, 0)
V2 (Front-Right)	(215.9, 0)
V3 (Right-Rear)	(215.9, 215.9)
V4 (Apex/Back)	(0, 431.8)
V5 (Left-Rear)	(-215.9, 215.9)

Construction - Sandwich Frame

The device uses a bolt-together sandwich frame: two CNC-milled aluminum plates (top & bottom) with polycarbonate panels sliding into perimeter channels, tied together by 5 M8 stainless steel through-rods at each vertex.

Panel Assignments

Panel	Width	Description
A (Front)	431.8 mm (17")	Faces pitcher, has display, cameras
B (Right)	215.9 mm (8.5")	Strike zone panel
C (Right-Diag)	304.8 mm (12")	Strike zone panel
D (Left-Diag)	304.8 mm (12")	Strike zone panel
E (Left)	215.9 mm (8.5")	Strike zone panel

Adjustable Leg System

Five telescoping legs with quick-detent pins allow height adjustment without tools. Each leg has a linear potentiometer that reports its height to the microcontroller, enabling automatic strike zone calibration.

Setting	Extension	Total Height	Target User
Low (Little League)	50.8 mm (2")	24"+2" = 26"	Ages 8-12, under 5'
Medium (High School)	127 mm (5")	24"+5" = 29"	Ages 13-18, 5'-6'
High (College/Pro)	203 mm (8")	24"+8" = 32"	6', extended zone

3. Engineering Analysis

Impact Force Analysis

A regulation baseball at 105 mph delivers approximately 160 Joules of kinetic energy. The peak impact force depends on surface compliance:

Surface	Contact Time	Peak Force
Rigid concrete	0.5 ms	12,920 N (2,904 lbf)
Wooden backstop	1.0 ms	6,460 N (1,452 lbf)
3/8" Polycarbonate	2.5-3.0 ms	2,153-2,584 N (484-581 lbf)
1/2" Polycarbonate	3.0-4.0 ms	1,705-2,154 N

Polycarbonate vs. Acrylic (Plexiglass)

WARNING: NEVER use acrylic/plexiglass. A 100mph baseball WILL shatter acrylic into dangerous shards. Polycarbonate (Lexan) is 37x more impact resistant and flexes to absorb the hit.

Property	Acrylic	Polycarbonate	Ratio
Impact Strength (Izod)	0.4 ft-lb/in	15+ ft-lb/in	37x
Elongation at Break	4%	110%	27x
Behavior at 100mph	SHATTERS	Flexes, recovers	CRITICAL
Cost per sq ft	~\$5	~\$8	1.6x

Structural Integrity

3/8" polycarbonate panels at the widest span (17") experience a maximum bending stress of ~4,073 psi under a 105 mph impact, well below the 10,000 psi yield strength. Safety factor: 2.49x. The aluminum frame (6061-T6, 276 MPa yield) has a safety factor >10x. Panel retention in EPDM-gasketed channels is ensured by >25,000 N of through-rod clamping force vs. <2,600 N impact ejection force.

Fatigue life: Polycarbonate endurance limit is ~2,000 psi. Calculated cyclic stress is ~1,800 psi per impact. Estimated panel lifespan: 500,000+ impacts (14+ years at 100/day).

Thermal Analysis

Under typical operation (LEDs at 30%, compute active), total heat dissipation is ~95W. With natural convection through top-plate ventilation slots and the aluminum top plate acting as a heatsink, the enclosure temperature rises only 13 deg C above ambient. Maximum safe ambient temperature: 45 deg C (113 deg F). Brief LED bursts (strike/ball effects) add 252W peak but last only 200ms, causing negligible heating.

Sensor Accuracy

Impact Location (Acoustic Triangulation)

Each panel has 4 piezo film sensors at its corners. Sound travels through polycarbonate at 2,200 m/s. The ESP32-S3 timestamps detections with +/-2 microsecond resolution. With 4-sensor trilateration, impact location accuracy is +/-1 cm. This enables the "illuminate precisely where the ball touches" feature.

Radar (IWR6843 60GHz FMCW)

Parameter	Specification
Speed Accuracy	+/- 1 mph at 60 feet
Range Accuracy	+/- 5 cm
Update Rate	50 Hz (20ms per reading)
Spin Rate Detection	Yes, via micro-Doppler analysis
Pitch Movement	Tracks lateral/vertical deviation from expected path
Effective Range	0.2m to 20m (covers 60'6" pitcher distance)

5. Pitching Analytics: Spin Rate & Movement

The SmartHome Plate provides professional-grade pitch analytics that were previously only available in major league stadiums with \$100K+ tracking systems. Using the 60GHz FMCW radar and the dual-camera system, the device measures:

Spin Rate Detection

Spin rate is measured using micro-Doppler analysis from the FMCW radar. A spinning baseball creates sidebands in the radar return signal. The frequency offset of these sidebands is directly proportional to the spin rate:

$$f_{\text{spin}} = (2 * \omega * r * \cos(\theta)) / \lambda$$

where ω = angular velocity, r = ball radius (~37mm), θ = spin axis angle

Pitch Type	Typical Spin	Movement Profile
4-Seam Fastball	1500-2500 RPM	Backspin, rises (less drop)
2-Seam Fastball	1200-1800 RPM	Backspin + arm-side run
Curveball	2000-3000 RPM	Topspin, 12-6 drop
Slider	1800-2600 RPM	Gyro spin, lateral sweep
Changeup	800-1500 RPM	Reduced spin mimics slower pitch
Knuckleball	0-300 RPM	Near-zero spin, unpredictable dance
Cutter	2000-2500 RPM	Backspin + glove-side break
Splitter	600-1200 RPM	Reduced spin, tumbling drop

Pitch Movement Analysis

Movement is defined as the deviation from an expected straight-line path from the pitcher to the plate. The radar tracks the ball's position at regular intervals during its flight, and the camera system confirms the trajectory:

How Movement is Calculated

- Radar tracks the ball's 3D position at 50 Hz during flight (every ~1.3 feet)
- Expected path: straight line from release point to center of zone
- Horizontal movement (induced break): deviation perpendicular to the straight path
- Vertical movement (drop/rise): deviation from gravitational drop line
- Total movement: vector magnitude of horizontal + vertical deviation

Pitch Type	Typical Movement	Description
4-Seam Fastball	5-12"	Rise effect (less than expected drop)
Curveball	8-15"	12-6 or 1-7 drop (depending on grip)
Slider	4-10"	Lateral sweep, late break
Changeup	2-6"	Arm-side fade, tumbling action
Knuckleball	3-8"	Unpredictable, changes direction multiple times

Knuckleball tracking: The radar samples at 50Hz and can detect the characteristic micro-movements of a knuckleball. The 3D trajectory plot shows the ball's sinusoidal or chaotic path, with each directional change counted and displayed. The "wobble index" quantifies overall unpredictability on a 0-10 scale.

6. Hitting Analytics: Exit Velocity & Launch Angle

The SmartHome Plate is dual-purpose. In hitting mode, the device is placed in the batter's box to measure ball flight characteristics and bat performance:

Exit Velocity

Exit velocity (EV) is the speed of the ball immediately after contact with the bat. The 60GHz FMCW radar tracks the ball's approach speed and, upon detecting a direction reversal (ball bouncing off), its departure speed:

- Approach speed: Pitch velocity (40-110 mph range)
- Exit velocity: Ball speed after bat contact (up to 120+ mph)
- Ratio calculation: Exit vel / Approach vel = efficiency metric
- Display updates in real-time (< 200ms) on the front display and app
- Historical tracking: Session averages, personal bests, trend analysis

Launch Angle

Launch angle is the vertical angle at which the ball leaves the bat, measured in degrees from horizontal. This is critical for understanding batted-ball outcomes:

Type	Angle Range	Outcome
Ground ball	< 10 degrees	High outs, low AVG
Line drive	10-25 degrees	Highest AVG, most productive
Fly ball	25-50 degrees	Home run range (25-35 optimal)
Pop up	> 50 degrees	Near-automatic outs

Launch angle is calculated from the radar's measured departure vector and camera trajectory analysis. The system captures the first 30 feet of ball flight post-contact to calculate the precise angle.

Batter Swing Analysis

The wide-angle camera captures the batter's swing from the side view:

- Swing plane angle: Measured from shoulder-to-hand path at contact point
- Bat speed: Estimated from video analysis of swing duration
- Late swing detection: If bat angle at contact is > 5 degrees behind the ball, display "LATE" alert
- Early swing detection: If bat angle is > 5 degrees ahead, display "EARLY" alert
- Contact point quality: Combines exit velocity, launch angle, and contact location

Dual Mode Summary

The device seamlessly switches between PITCHING MODE and HITTING MODE based on its placement. When placed on home plate, it enters pitching mode (strike zone, pitch analytics). When placed in the batter's box, it enters hitting mode (exit velocity, launch angle, swing analysis).

7. Auto-Adjusting Strike Zone

A fixed-height device cannot serve a 10-year-old Little Leaguer and a 6'3" college pitcher. The SmartHome Plate solves this with a three-tier adaptation system:

1. Physical: Adjustable Telescoping Legs

Five quick-detent legs provide three height positions that set the physical device to approximately match the batter's zone:

- LOW (2" extension): 26" total height - Little League, ages 8-12
- MEDIUM (5" extension): 29" total height - High School, ages 13-18
- HIGH (8" extension): 32" total height - College/Pro, 6' and above

Each leg includes a linear potentiometer that reports its extension to the microcontroller.

2. Software: LED-Illuminated Zone Boundaries

Regardless of the physical height, the LED matrix can illuminate only the rows that correspond to the current batter's strike zone. The panel has ~4,200 NeoPixels in a grid; each "row" of LEDs corresponds to a 1-inch vertical band. The active zone illuminates with a bright blue border:

- Zone bottom (knees): Blue line + slight glow
- Zone top (mid-torso): Blue line + slight glow
- Between: Subtle ambient glow marking the valid zone
- Outside zone: Panels dim but still detect impacts for ball/strike calls

3. A

The wide-angle camera uses ML pose estimation (MoveNet Thunder running on the Jetson) to automatically detect the batter and calculate their strike zone:

Detection Process

1. User presses "ZONE SET" button or triggers via app
2. Camera captures the batter standing next to home plate
3. MoveNet Thunder detects 17 body keypoints (knees, hips, shoulders)
4. Strike zone bottom = midpoint of left & right knee heights
5. Strike zone top = midpoint between shoulders and hips
6. System maps zone to LED row numbers and illuminates boundaries
7. User confirms zone via app or green button press
8. Zone locks in for the current session or until re-detection

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8. Game Modes

Mode	Description	Scoring
Practice	Free pitch. All stats displayed. Adjustable targets. Session recording.	Session stats, spin/movement per pitch
Simon Says / Pop-a-Shot	One zone of one panel lights up. Pitcher must hit it within 3 seconds. Targets get points and fractional leaderboard by league round.	Points per hit, fractional leaderboard by league round
Game Mode	1-20 innings, 1-10 pitchers/teams. Full ball/strike calling. Simulates a real game.	Fewest balls = winning team
Batter's Box	Device in batter's box. Calls balls/strikes on real pitches during batting practice.	Pitch accuracy stats for the pitcher
Hitting Mode	Device in batter's box. Measures exit velocity, launch angle, contact quality.	Exit velocity averages, launch angle distribution, swing

Anti-Cheat Detection

For competitive game modes (Simon Says, Pop-a-Shot), the system validates every target hit:

- Impact must register on the ILLUMINATED panel, not a different one
- Impact force must exceed minimum threshold (equivalent to >20 mph pitch)
- Radar must confirm an approaching object exceeding 30 mph (prevents tapping the device)
- Impact timestamp must be within 3 seconds of target appearing
- Repeated violations trigger a "suspected interference" warning and 30-second cooldown

League Divisions (Pop-a-Shot)

League	Target Size	Speed Requirement	Leaderboard
Little League	Target = 50% of panel width	No spin requirement	Top 100 nationally
High School	Target = 35% of panel width	30+ mph required	Top 100 nationally
College	Target = 25% of panel width	50+ mph required	Top 100 nationally
Pro	Target = 15% of panel width	70+ mph required	Top 100 nationally

9. Bill of Materials & Cost Analysis

Material Cost Breakdown

Component	Cost
Polycarbonate panels (5x)	\$45
Aluminum plates (top+base, CNC)	\$85
Aluminum extrusions (legs+rods)	\$30
Hardware (fasteners, gaskets, feet)	\$25
Jetson Orin Nano (Pro model)	\$250
ESP32-S3 (5x)	\$25
Wide-angle camera	\$35
High-speed camera (OV9281)	\$25
Radar module (IWR6843ISK)	\$55
5" IPS display	\$40
WS2812B LEDs (~4,200)	\$60
Piezo sensors (20x)	\$15
Battery pack (6x 21700)	\$50
3D printed parts (PETG)	\$35
Misc (wiring, PCBs, connectors)	\$40

TOTAL MATERIALS (Pro model)

\$815

Assembly labor (2 hours)

\$40

QC & Testing (1 hour)

\$20

TOTAL COGS (Pro)

\$875

Pricing Recommendation

Configuration	COGS	MSRP	Margin	Target Market
Standard (RPi CM4)	\$750	\$899	1.2x	Budget-minded users, Little League
Pro (Jetson Orin)	\$875	\$1,299	1.48x	Competitive pitchers, hitting training
Pro+ (Motorized legs)	\$1,050	\$1,499	1.43x	Pro facilities, automated zone detection

At \$1,299 MSRP for the Pro model, the device offers significant value compared to dedicated radar guns (\$300-600), pitching trainers (\$500-2,000), and tracking systems that cost \$5,000+. The SmartHome Plate combines ALL of these functions in one device.

10. CNC Manufacturing Files

Seven CNC-ready DXF files are included in this package, located in the `cnc/` directory. Each file contains manufacturing layers:

File	Part	Material	Process
base_plate.dxf	Base plate	6061-T6 Aluminum, 1/4"	3-axis CNC mill
top_plate.dxf	Top plate	6061-T6 Aluminum, 1/4"	3-axis CNC mill
front_panel.dxf	Front panel (17")	Polycarbonate, 3/8"	CNC router
side_panel_left.dxf	Left side (8.5")	Polycarbonate, 3/8"	CNC router
side_panel_right.dxf	Right side (8.5")	Polycarbonate, 3/8"	CNC router
rear_panel_left.dxf	Left diagonal (12")	Polycarbonate, 3/8"	CNC router
rear_panel_right.dxf	Right diagonal (12")	Polycarbonate, 3/8"	CNC router

DXF Layer Guide

Layer	Content	Use
CUT (Red)	Cut profiles and through-holes	Send to machine
REFS (Green)	Reference marks, sensor positions	Assembly aids only
CENTER (Blue)	Hole center marks	Drill start points
DIMS (Yellow)	Dimensions and notes	Reference only
ENGRAVE (White)	Engraved text/lines	Optional decorative

3D-Printed Parts

Nine 3D-printed parts in PETG are required. OpenSCAD source files are in `cad/parts/`. Export to STL using OpenSCAD (File > Export > STL):

Part	Settings	Qty	Time
Electronics tray	PETG, 30% infill	1	8 hours
Camera housing (wide)	PETG, 40% infill	1	2 hours
Camera housing (high-speed)	PETG, 40% infill	1	2 hours
Display mount frame	PETG, 30% infill	1	4 hours
LED panel frames (5x)	PETG, 20% infill	5	6 hours each
Battery sled	PETG, 40% infill	1	3 hours
Corner cable guides	PETG, 100% infill	5	1 hour each
Radar module mount	PETG, 40% infill	1	1 hour
BMS board mount	PETG, 40% infill	1	1 hour

11. Assembly Guide (16 Steps)

Step 1: CNC Mill Aluminum Plates

Mill top and base plates from 1/4" 6061-T6 aluminum per DXF drawings. Deburr all edges, break corners. Anodize optional.

Step 2: CNC Route Polycarbonate Panels

Route all 5 panels from 3/8" UV-hardcoated Lexan per DXF drawings. Keep protective film on outer surface.

Step 3: Install Heat-Set Brass Inserts

M4 (8x) in top plate, M3 (16x) in 3D printed parts. Use soldering iron at 230C, press and hold 5s.

Step 4: Assemble Adjustable Feet

Slide inner tube into outer tube. Align detent holes. Insert spring pins. Screw rubber feet onto M6 studs.

Step 5: Install EPDM Gaskets

Cut gasket lengths for all 5 channel segments. Press into base and top plate channels. Apply silicone at joints. Cure 24h.

Step 6: Install Polycarbonate Panels

Place base plate flat. Slide bottom edge of each panel into base channel. Work front-to-back (A, B, C, D, E).

Step 7: Install Top Plate

Carefully lower top plate over panel top edges. Align all 5 channels simultaneously. Press down firmly.

Step 8: Install Through-Rods

Insert 5x M8 threaded rods through all vertices. Add washers and nuts top and bottom. Torque to 8 Nm in star pattern.

Step 9: Install LED Matrix Strips

Clean panel inner surfaces with IPA. Apply WS2812B strips horizontally every 33mm. Solder daisy chains.

Step 10: Install Piezo Sensors

Adhere 4x LDT0-028K sensors per panel at corners. Route leads through cable guides to ESP32 boards.

Step 11: Mount Electronics

Shock-mount tray to top plate with M4 screws through silicone dampeners. Install Jetson, ESP32 boards, BMS, buck converters.

Step 12: Install Cameras & Radar

Mount cameras in top plate housings. Route FPC cables. Mount radar module. Apply anti-fog coating to lens windows.

Step 13: Install Display

Slide display into mount frame. Connect HDMI + power. Snap frame into front panel cutout. Secure with M3 screws.

Step 14: Connect Battery

Install 6x 21700 cells in sled. Connect BMS leads. VERIFY POLARITY with multimeter. Slide sled into base plate cavity.

Step 15: Final Wiring & Sealing

Route all cables through cable guides. Apply silicone sealant around cameras, display, cable glands. Cure 24h.

Step 16: Software & Calibration

Flash Jetson with Ubuntu + drivers. Flash ESP32s with firmware. Configure WiFi. Run calibration: IMU level, radar distance, piezo tap test, LED rainbow test, camera focus.

12. Firmware Architecture

The system uses a dual-processor architecture: a Jetson Orin Nano for ML/video/radar processing, and 5 ESP32-S3 microcontrollers for real-time sensor polling and LED control.

Jetson Orin Nano (Main Compute)

- Camera pipeline: GStreamer -> 4K wide-angle @30fps + 120fps global shutter
- ML pipeline: TensorRT -> MoveNet Thunder pose estimation + pitch type classifier
- Radar pipeline: TI mmWave SDK -> Doppler analysis -> speed + spin + movement
- Video: 2-second ring buffer, triggered by radar detection (>30mph approach)
- Comms: WiFi 6 + BT 5.2 for app connection

ESP32-S3 (Per-Panel Controllers)

- Impact detection: 4-channel ADC at 10 kHz per sensor (20 sensors total)
- Acoustic triangulation: +/-1 cm accuracy, <500us response time
- LED control: WS2812B protocol, 800+ LEDs per ESP32, daisy-chained data
- Communication: UART daisy-chain to master ESP32 -> Jetson

Impact Detection Algorithm

- Continuous ADC sampling at 10 kHz on all 4 channels per panel
- Bandpass filter: 100 Hz - 5 kHz (remove ambient noise)
- Threshold crossing: >5g equivalent (~800 ADC counts)
- On detection: capture 500 samples, timestamp each sensor crossing
- Trilateration: solve for (x,y) impact point using differential timing
- Force estimation: sum of all 4 sensor amplitudes, calibrated to Newtons
- Result sent to Jetson in <1ms: {panel, x, y, force, timestamp}

13. Companion App Architecture

Minimum Viable Product (Web App - Phase 1)

The first version will be a Progressive Web App (PWA) that works on any device with a browser. This reaches the widest audience with the fastest development time. Native iOS/Android apps follow in Phase 2.

Web App Tech Stack

- Frontend: React + TypeScript + Tailwind CSS (PWA with offline support)
- Backend: Node.js + Express + WebSocket (for real-time data)
- Database: PostgreSQL for user data, InfluxDB for time-series pitch data
- Real-time: WebSocket for pitch-by-pitch updates
- Video: WebRTC for streaming from device, HLS for recorded sessions
- Deployment: Vercel (frontend) + Railway (backend)

Core Screens

Screen	Description	Priority
Dashboard	Live session view, current stats, real-time pitch data	P0
Session Review	Video replay with overlaid stats (speed, spin, movement)	P0
Historical Stats	All-time stats, trends, comparison charts	P0
Session Setup	Select mode, configure zones, set targets	P0
Leaderboard	Pop-a-Shot scores by league, weekly/monthly/all-time	P1
Game Mode	Innings, pitch count, balls/strikes, teams	P1
Hitting Mode	Exit velocity, launch angle, contact quality	P1
Device Settings	WiFi config, calibration, firmware updates	P0

Data Flow

Device -> WiFi -> Cloud -> App: Each pitch generates a data packet containing: {timestamp, speed_mph, spin_rpm, movement_horiz_in, movement_vert_in, pitch_type_estimate, strike_zone_x, strike_zone_y, impact_force, result (ball/strike)}. Video is stored locally on device and synced to app on request.

14. Safety Warnings & Maintenance

WARNING: CRITICAL: This device is designed to withstand repeated 105+ mph baseball impacts. However, always stand behind a protective screen when using it with live pitching. Never stand directly in the line of fire.

WARNING: MATERIAL SAFETY: Only use polycarbonate (Lexan) for panels. Acrylic (Plexiglass) WILL shatter and create dangerous flying shards. Verify panel material before each use.

WARNING: ELECTRICAL SAFETY: This device contains a 111Wh lithium battery. Do not expose to water beyond IP54 rating. Do not modify or puncture battery. Use only the included charger.

Maintenance Schedule

Interval	Task
Every use	Wipe panels with microfiber cloth. Check for loose fasteners.
Weekly	Clean camera lenses. Verify battery charge. Check LED function.
Monthly	Run full diagnostic (all sensors, all LEDs, radar, cameras). Update firmware.
Quarterly	Inspect EPDM gaskets for wear. Replace silica gel desiccant pack.
Annually	Full disassembly inspection. Replace any cracked panels. Re-calibrate radar.
As needed	Replace worn rubber feet. Clean ventilation slots. Reset device.